

Lesson 11 — Build the Model & Repeat It

Matches LEGO Teacher Portal lesson: "Gold Patrol".

Objective (I can...): I can build the loops model and use a repeat loop so I don't copy the same block many times. **CS concept:** Loops / repetition (repeat N times). A loop is **abstraction — hiding the messy details**: one "repeat 5x" block stands in for five copied blocks. · **Builds on:** Lesson 10 (events start actions we now repeat).

AI Connection (2 min): AI gets better by repeating — training means showing examples over and over. *First-principles question:* "Why would more tries help a computer learn?"

Materials: LEGO CS & AI kit + device per pair. **This is the unit's build lesson** — build the Loops model now and keep it assembled through Lesson 15.

60-minute flow - Warm-up (8): "Do it again" — clap 10 times, then find a shorter instruction. Reveal: "repeat 10 times" is a loop. - **Concept + demo (10):** Show 4 copied move blocks, then replace them with one move inside a repeat-4 loop; run both to show they match. - **Build/Code (30):** Pairs **build the Loops model** (the unit's one build), then program a repeated action with a repeat loop (spin, pause, repeat 5x). *Swap at 15 min.* - **Challenge (7):** Change the repeat number; predict how the behavior length changes before running. - **Share + pack-up (5):** Pairs show their loop and state the count. **Leave the model built.**

Pair roles: Driver builds/codes, Navigator counts the repeats aloud. Swap at 15 min. (*Trio rotates.*)

Differentiation: *Support* — a labeled palette card highlighting the repeat block. *Extension / fast-finisher* — put two blocks inside the loop. *Reteach* — count the repeats out loud together on the first run.

Exit check: Pair replaces repeated blocks with a working repeat loop and can state the count. **At-home (optional):** Find something you do "over and over"; write it as "repeat ___ times." ~10 min.
